Def. Define

$$f_a^c : [-1,1] \to \mathbb{R} \; ; \; x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x^a \cdot \sin(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$

where $a \in \mathbb{R}$, $c > 0$. And, define quotient at $x=0$ of $f_a^c$ as

$$q_a^c : [-1,1] \setminus \{0\} \to \mathbb{R} \; ; \; t \mapsto \frac{f_a^c(t)}{t}$$

Then, $q_a^c(t) = t^{a-1} \cdot \sin(|t|^{-c})$.

Prop. $\frac{d}{dx} f_a^c(x) = \begin{cases} \lim_{t \to 0} t^{a-1} \cdot \sin(|t|^{-c}) & \text{if } x=0, \text{ and the limit exists.} \\ a \cdot x^{a-1} \cdot \sin(|x|^{-c}) - c \cdot x^{a+1} \cdot |x|^{-c-2} \cdot \cos(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$

Simplify $a x^{a-1} \cdot \sin\left(\frac{1}{x^c}\right) - c \cdot x^{a-c-1} \cdot \cos\left(\frac{1}{x^c}\right)$ if $x > 0$

$$2x \sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right)$$

$$\lim_{t \to 0} t \cdot \sin$$

$$x^a \cos(|x|^{-c}) \cdot -c \cdot x^{-c-1}$$

$$+ x^a \cdot \cos(|x|^{-c}) \cdot \frac{x c}{|x c|} \cdot (-c) \cdot |x|^{-c-1}.$$

| Equivalent Condition Table | $f$ | $f'$ | $f''$ |
|---|---|---|---|
| Continuous | $a > 0$ | $a > 1+c$ | $a > 2+2c$ |
| bounded | $a \geq 0$ | $a \geq 1+c$ | $a \geq 2+2c$ |
| Exists at $x=0$ | Always. | $a > 1$ | $a > 2+c$ |
| Bounded Variation | | | |

[f]

Observe that $f:[-1,1]\to\mathbb{R};\ x\mapsto \begin{cases} 0 & \text{if } x=0 \\ x\sin\left(\frac{1}{|x|}\right) & \text{if } x\neq 0 \end{cases}$.

Then, f is Continuous, Not differentiable at $x=0$, Not bounded variation, Not bounded derivative,

proof. since $\left|x\sin\left(\frac{1}{|x|}\right)\right| \le |x| \to 0$ as $x\to 0$, Clearly continuous.

Meanwhile, $f'(0)$ exists $\Leftrightarrow \lim_{t\to 0}\sin\left(\frac{1}{|t|}\right)$ exists, this not exists.

And, to show that f is Not bounded variation, Construct a Partition as:

$$P_n \overset{\text{def}}{=} \left\{-1, 0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \cdots, \frac{2}{2\pi}, \frac{2}{\pi}, 1\right\}$$

Then, $V(f, P_{2n+1}) \ge \sum_{k=0}^{n-1} \frac{2}{(2k+1)\pi} = \frac{2}{\pi}\cdot\sum_{k=0}^{n-1}\frac{1}{2k+1} \to \infty$ as $n\to\infty$.

Therefore it is Not bounded variation. Similarly, $f_n^n$ $(n\in\mathbb{N})$ are Not bounded variation.

But, in this way, we can not determine $f_2'(x) = x^2\sin\left(\frac{1}{x}\right)$ is BV or Not.

Observe that

$$\frac{d}{dx}f_2'(x) = \begin{cases} 0 & \text{if } x=0 \\ 2x\sin\left(\frac{1}{x}\right) - \cos\left(\frac{1}{x}\right) & \text{if } x>0 \end{cases}$$

It is Riemann integrable and bounded. Therefore, for any partition P on $[-1,1]$,

$$V(f_2', P) = \sum_{i=1}^{n}|f(x_i) - f(x_{i-1})| = \sum_{i=1}^{n}|f(t_i)|\cdot\Delta x_i \le \sum_{i=1}^{n}\sup|f'|\cdot\Delta x_i = \sup|f'|\cdot 2$$

(MVT)

Therefore, $f_2'$ is Bounded Variation. Corollary,

[Prop. If $f:[a,b]\to\mathbb{R}$ is differentiable and the derivative $f'$ is bounded, then f is bounded variation.

proof. Ask a toddler on the street.

Prop. $f_2^2$ is not bounded variation.

Pf. Let $P_n = \{x_1, x_2, \cdots, x_n, x_{n+1}\}$ as $x_1 = -1$, $x_{n+1} = 1$, and

$$x_k = \frac{1}{\sqrt{\left(\frac{1}{2}+n-k\right)\pi}} \qquad 0 \le k \le n$$

$$\frac{1}{x^2} = \left(\frac{1}{2}+n\right)\pi$$

$$x = \frac{1}{\sqrt{\left(\frac{1}{2}+n\right)\pi}}$$

Then, $V(f_2^2, P_n) \ge \sum_{i=1}^{n} \left| f_2^2(x_i) - f_2^2(x_{i-1}) \right|$

$$= \sum_{i=1}^{n} \frac{1}{\left(\frac{1}{2}+n-i\right)\pi} + \frac{1}{\left(\frac{1}{2}+n-(i-1)\right)\pi}$$

$$\ge 2 \cdot \sum_{i=1}^{n} \frac{1}{\left(\frac{1}{2}+n-i+1\right)\pi}$$

$$= \frac{2}{\pi} \cdot \sum_{i=1}^{n} \frac{1}{\frac{1}{2}+i} \ge \frac{2}{\pi} \sum_{i=1}^{n} \frac{1}{i} \to \infty \quad \text{as } n \to \infty$$